Robotic's Ethics and Applications

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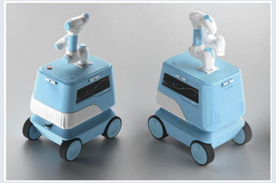
What is Robotics?

Robotics entails the operation, design, and manufacturing of robots which are used to complete tasks in multiple different settings. [1]

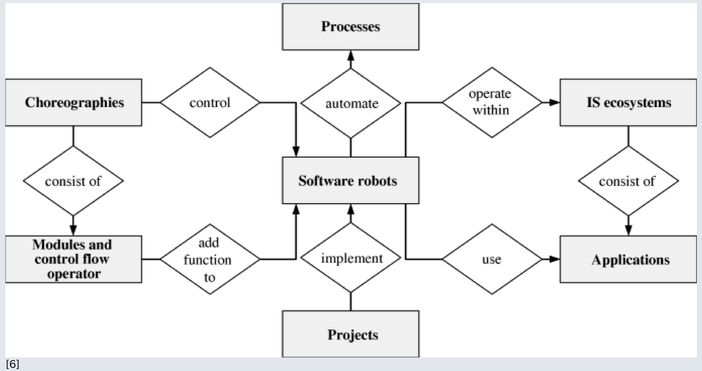


[2]

Robots have revolutionized entertainment, healthcare, transportation industries. Robotics has grown due to advances in AI, sensor technology, and machine learning. [1]

Robotic Process Automation (RPA)

- RPA is software that has set rules and defined capabilities that allow it to automatically complete tasks in separate software systems. [6], [7]
- RPA is useful for completing repetitive tasks and it is effective at data management, integrating systems, and overall improving processes. [6]
- RPA can be used in a multitude of different ways and can be used to improve areas such as security, auditability, scalability, convenience, and compliance. [6]



[6]

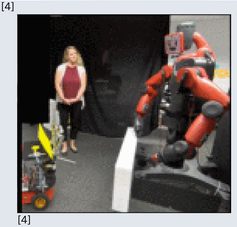
- There are 2 types of RPA, assisted and unassisted RPA. [7]
- Assisted RPA requires human assistance in order to perform tasks. [7]
- Unassisted RPA functions without the need for humans to initiate, monitor, or end the process upon completion. Unassisted RPA can function 24/7 and send alerts if something goes wrong. [7]

The Human Component

Safety

- The rise of robotics in our daily lives, coupled with our increased reliance on automation for industrial production, has spawned an industry completely based around robotic safety. Safety can be considered from both a physical and ethical perspective. [3]
- The physical component focuses on preventing workplace injury. [3] This was most prevalent during the Industrial Revolution when machinery was extremely unsafe. Contemporary history has fixed most of these issues.

[4]



[4]

Ethics

- It's important to consider the ethical consequences of increased usage and reliance on robotics. Humans' jobs are being lost due to an increase in the usage of robotics. [5]
- In the future, there may be very few skills that machines can't perform better than humans. [5]
- There is also the issue of how automation will respond to ethical dilemmas. For example, if an autonomous car must choose between crashing (possibly injuring or killing its passenger) or running over a pedestrian, what choice should the car make? Who is culpable if the choice the car makes is proven wrong? [5]

Industries where RPA's are used

Healthcare & Pharmaceutics

RPA could be used in healthcare to help speed up the process of identifying and analyzing health conditions. It can also improve productivity and work quality as well as reduce costs. Automation has allowed for some complicated tasks in pharmaceutics to be completed without human intervention. [7]

Accounting & Finance

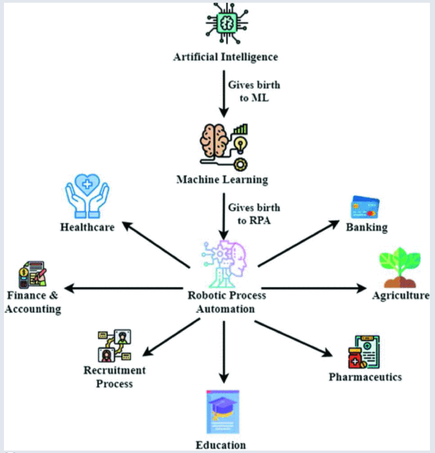
Automation is used to reduce costs and improve data security as well as compliance processes. It can be used to process invoices, manage payments, cross-check purchases, and respond to questions from vendors. [7]

Education

Automation is used to aid in enrollment processes, and scheduling of meetings as well as to reduce repetitive work and paper consumption. [7]

Banking

In banking, automation manages credit cards, mortgages, account payments, account closures, and customer services. It also aids in fraud detection. There is more productivity in employees through the use of automation to answer customers' questions. [7]



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